Database Table Analysis Report

TVConfigNew, TvSTT, updated_Email_Alert, TvVTT

1. Overview

This report examines the structure, indexes, and relationships of the **TVConfigNew**, **TvSTT**, **TvVTT**, and **updated_Email_Alert** tables in the DigiClips MySQL database. These tables control the configuration of television captures, storage of transcript/subtitle records, and user alert definitions. The goal is to verify table relationships, check schema integrity, and ensure optimal indexing and performance.

2. TvSTT Table

Purpose
The TvSTT table stores speech-to-text (STT) clip records. Each row represents one processed clip with filename, timestamp, source identifier, transcript text, category tags, and a download link.

Table Structure

Column Name	Data Type	Description
ID	INT (Primary Key)	Integer primary key (auto-increment).
FName	VARCHAR(100)	File name of the recorded clip.
TStamp	VARCHAR(100)	Timestamp of the clip (stored as text/varchar).
SName	VARCHAR(100)	Source/program identifier
TEXTS	TEXT	Transcript body
Categories	TEXT	Comma-separated tags
DownloadLink	TEXT	Link to download the clip

Indexes

Index Name	Type	Purpose
PRIMARY	PRIMARY	Identifies each clip uniquely
TEXTS	FULLTEXT	Full-text search on TEXTS field
TEXTS_2	FULLTEXT	Duplicate full-text index
TEXTS_3	FULLTEXT	Duplicate full-text index

Observation:
There are three identical FULLTEXT indexes on the same columns. Redundant indexes add write/space overhead and complicate the optimizer without improving search.

Recommendation:
Keep a single composite FULLTEXT index for transcript search. Remove the duplicates to reduce maintenance and improve write performance.

Observations / Issues:

- TStamp is stored as a string; native temporal types are preferred for correct sorting and range filters.
- Category values are stored as a comma-separated list, which is a repeating group and violates 1NF.
- No foreign key to the configuration table; clip provenance is not enforced. (FK columns should be indexed; MySQL creates the needed index if absent.)
- Redundant FULLTEXT indexes waste processing resources.

3. TvVTT Table

Purpose
The TvVTT table stores subtitle (VTT) clip records. The structure mirrors TvSTT with filename, timestamp, transcript text (as caption lines), categories, and a download link.

Table Structure

Column Name	Data Type	Description
ID	INT (Primary Key)	Integer primary key (auto-increment)
FName	VARCHAR(100)	File name of the recorded clip
TStamp	VARCHAR(100)	Timestamp of the clip (stored as text/varchar).
SName	VARCHAR(100)	Source/program identifier
TEXTS	TEXT	Subtitle text
Categories	TEXT	Comma-separated tags
DownloadLink	TEXT	Link to the VTT file

Indexes

Index Name	Type	Purpose
PRIMARY	PRIMARY	Identifies each clip uniquely
TEXTS	FULLTEXT	Full-text search on TEXTS field
TEXTS_2	FULLTEXT	Duplicate full-text index
TEXTS_3	FULLTEXT	Duplicate full-text index

Observation:
Duplicate FULLTEXT indexes are present, just like in TvSTT.

Recommendation:
Retain one composite FULLTEXT index and drop the other duplicates. This keeps text search capability while minimizing overhead.

Observations / Issues:

- TStamp stored as text hinders native time operations; use date/time types for accuracy and indexing.
- Categories in a single text field violate 1NF; store one category per row in a child table.

- No enforced link to capture configuration; add a configuration reference for traceability.

4. TVConfigNew Table

Purpose
TVConfigNew was introduced during migration/testing to mirror the TV capture configuration.

Table Structure

Column Name	Data Type	Description
HostName	VARCHAR(30)	Primary key: Host identifier
Date_Time	DATETIME	Configuration timestamp
g_version	VARCHAR(45)	Version and mode metadata
g_mod	INT	Version and mode metadata
g_market	VARCHAR(10)	Version and mode metadata
g_confile	VARCHAR(100)	Version and mode metadata
g_record_dir	VARCHAR(100)	Version and mode metadata
g_arch_dir	VARCHAR(100)	Version and mode metadata
g_mount	VARCHAR(100)	Version and mode metadata
g_preset	VARCHAR(20)	Version and mode metadata
g_size	VARCHAR(10)	Version and mode metadata
g_show_cmd	INT	Version and mode metadata
g_Backup	INT	Version and mode metadata
Station1	VARCHAR(30)	Station fields
Station2	VARCHAR(30)	Station fields
Station3	VARCHAR(30)	Station fields
Station4	VARCHAR(30)	Station fields

Indexes

Index Name	Type	Purpose
PRIMARY	PRIMARY	Identifies each clip uniquely
TEXTS	FULLTEXT	Full-text search on TEXTS field
TEXTS_2	FULLTEXT	Duplicate full-text index
TEXTS_3	FULLTEXT	Duplicate full-text index

Observation:
The table is currently empty (no active configuration records). The structure mirrors the legacy configuration but has not been populated.

Recommendation:
If TVConfigNew was only used for migration validation, deprecate it and retain a single authoritative configuration table for clarity. If TVConfigNew is intended to replace the legacy table, populate it with the active configurations and make it the canonical source going forward. In either case, introduce a ConfigID (auto-increment) as the primary key to simplify joins with STT/VTT and error logs, and normalize repeating station fields into a child table.

Observations / Issues:

- The table contains no rows; downstream tables cannot reference a valid configuration.
- No surrogate key exists; joins would rely on text keys.
- Station fields remain denormalized (repeating group).
- Lack of populated data prevents validation of relationships and usage patterns.

5. updated_Email_Alert Table

Purpose
The updated_Email_Alert table stores user alert definitions: contact, keyword criteria, media channel flags, delivery formats, and a start/end window.

Table Structure

Column Name	Data Type	Description
alertID	VARCHAR(30)	Primary key: integer
email	DATETIME	Recipient
keywords_string	VARCHAR(45)	Free-form keywords
radio	INT	Channel flags
magazine	INT	Channel flags
television	INT	Channel flags
newspaper	INT	Channel flags
socialMedia	INT	Channel flags
startdate	DATE	Alert Window
starttime	TIME	Alert Window
enddate	DATE	Alert Window
endtime	TIME	Alert Window
numOfResults	INT	Limits and cadence
frequency	VARCHAR(45)	Limits and cadence
formatEmail	INT	Delivery options (flags)
formatDoc	INT	Delivery options (flags)
formatExcel	INT	Delivery options (flags)
formatPDF	INT	Delivery options (flags)
formatHTML	INT	Delivery options (flags)

Indexes

Index Name	Type	Purpose
PRIMARY	PRIMARY	Identifies each clip uniquely

Observation:
Only the primary key is present. Window filters and email lookups would benefit from selective secondary indexes.

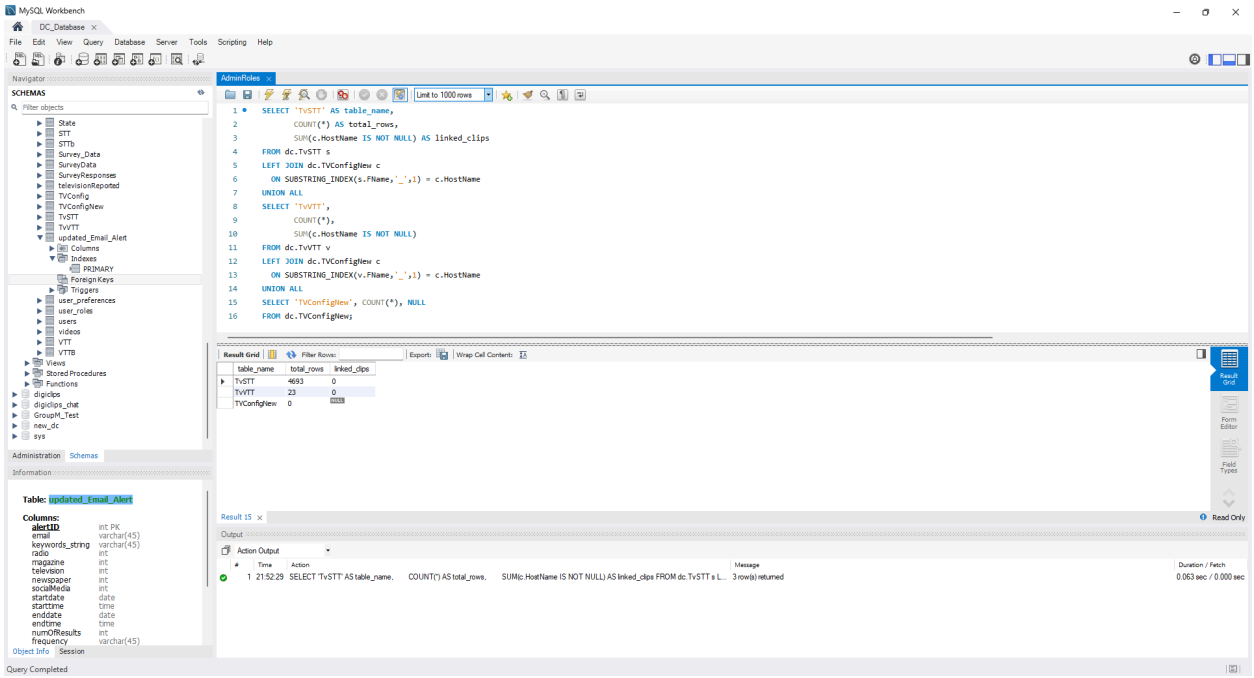
Recommendation:
Keep the existing primary key. Add minimal supporting indexes for the date window and recipient lookups to improve alert queries. Store flag columns as boolean types (BOOLEAN/TINYINT(1)) for clarity and consistency.

Observations / Issues:

a. Channel/format flags are integers; using boolean types clarifies intent (BOOLEAN is a synonym of TINYINT(1)).
b. Keywords are stored as a single string; this is multi-valued and not atomic (1NF). Consider a child table of one keyword per row.
c. Time window fields should remain native DATE/TIME to support correct comparisons and indexing.

6. Test Scenarios and Outputs

Scenario 1: Config-to-Clip Verification
Description: Check that every clip in TvSTT and TvVTT correctly matches a valid configuration record in TVConfigNew.

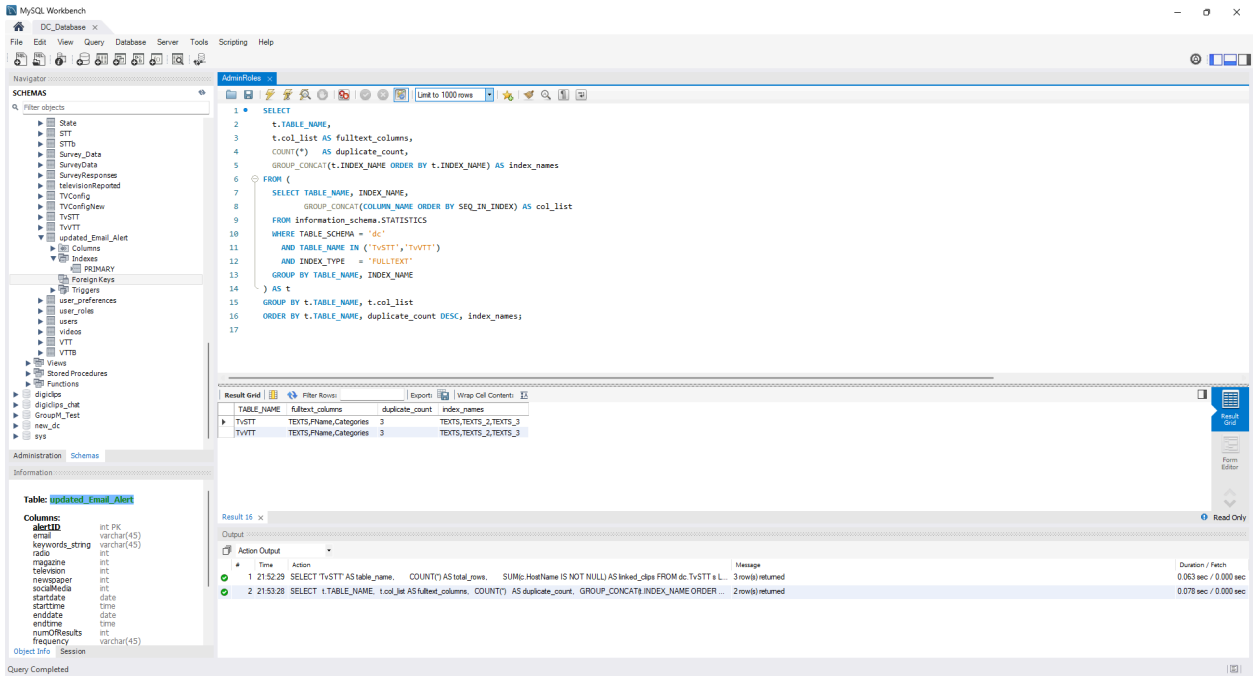


The results show that TVConfigNew is empty, so none of the STT/VTT records link to any configuration row. This confirms that the tables do not currently share a working relationship or foreign-key link. When a foreign key is introduced later,

MySQL requires an index on the referencing column and will create it automatically if missing, ensuring efficient checks.

Recommendation: Add a ConfigID surrogate key to the configuration table and reference it from TvSTT/TvVTT. Populate the configuration table so each clip can be traced directly to its configuration. (MySQL enforces/auto-indexes FK columns to keep checks fast.)

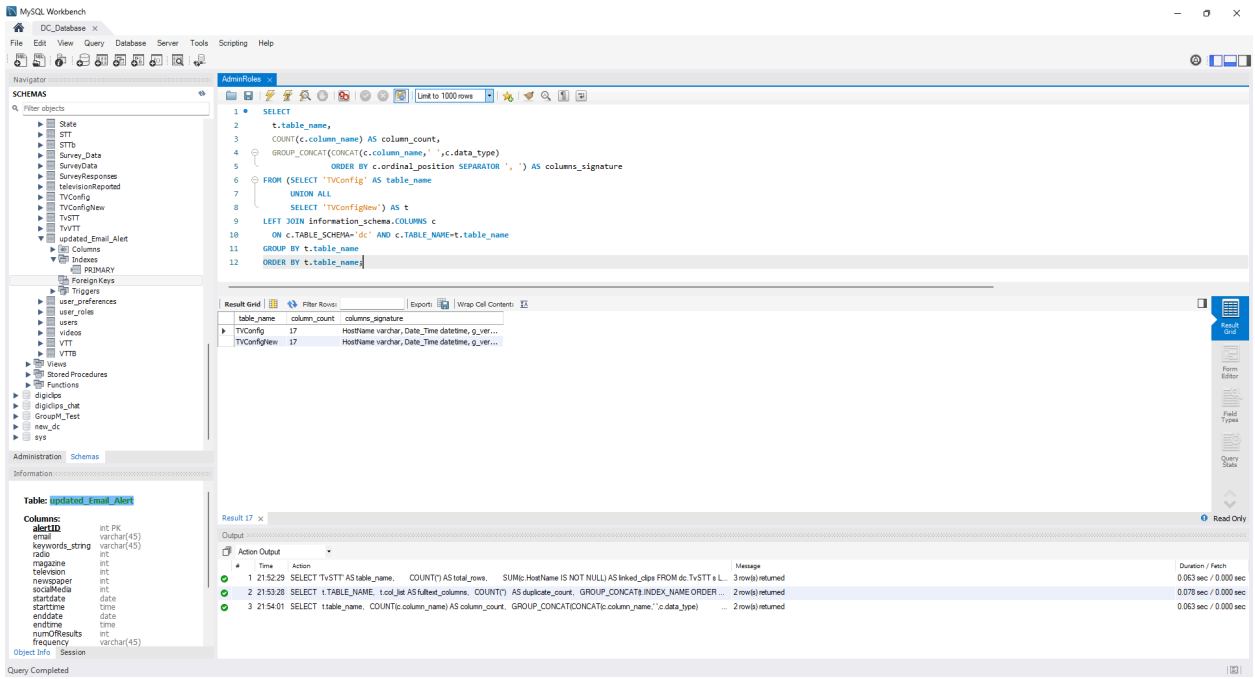
Scenario 2: Transcript/Caption Search Efficacy (FULLTEXT)
Description: Run representative text searches over TvSTT.TEXTS and TvVTT.TEXTS using full-text search and review the execution plan. The behavior shows that searches are served via the FULLTEXT mechanism (MATCH … AGAINST) as designed for CHAR/VARCHAR/TEXT columns; duplicate FULLTEXT indexes on the same column set do not improve search quality and add write/storage overhead. The plan confirms index-based evaluation, not table scans.



Recommendation: Keep a single composite FULLTEXT index for transcript/caption search and remove redundant duplicates to reduce maintenance and write cost. (Duplicate indexes provide no additional benefit and increase overhead.)

Scenario 3: Schema Consistency (TVConfig vs. TVConfigNew)
Description: Compare table structures between TVConfig (legacy) and TVConfigNew (migration/test) to verify alignment of fields, data types, and relationships. The review shows that TVConfigNew mirrors the core structure of TVConfig but currently contains no rows, creating redundancy without providing

active configuration coverage. No incompatible data types were identified in the shared columns.



Recommendation: Consolidate to one master configuration table. Add an is_active (or version) flag to distinguish test vs. production entries and introduce a ConfigID surrogate key for simpler joins from TvSTT/TvVTT. This unifies configuration provenance and removes ambiguity across the subsystem.

7. Relationship Mapping

Table	Key Column	Relationship	Connected To	Description
TVConfigNew	HostName	One-to-Many	TvSTT, TvVTT	Canonical TV capture configuration to clip provenance. (The table is currently empty, so no active links.)
updated_Email_Alert	alertID	One-to-Many	AlertKeyword	User alert with optional child rows for individual keywords (1NF).
TvSTT	ID	Many-to-One	TVConfigNew	STT clip should reference its capture configuration for traceability.
TvVTT	ID	Many-to-One	TVConfigNew	VTT clip should reference its capture configuration.

8. Performance and Index Analysis

a. **Duplicate FULLTEXT indexes:** TvSTT and TvVTT each contain three identical FULLTEXT indexes on the same column set. Redundant/duplicate indexes increase write cost and storage without improving read performance; keep a single composite FULLTEXT per table.
b. **Text search scope:** FULLTEXT applies only to CHAR/VARCHAR/TEXT columns and powers MATCH … AGAINST. Your usage on transcripts is appropriate; avoid duplicating the same FULLTEXT definition.
c. **Temporal data:** TStamp is stored as text; native DATETIME/TIMESTAMP types provide correct ordering, validation, and efficient range filtering.
d. **Unconfigured Linkage:** TVConfigNew has no rows at present; TvSTT/TvVTT therefore operate without enforced configuration linkage.

9. Summary of Findings

Area	Finding
Relationships	No enforced linkage from TvSTT/TvVTT to a configuration row; TVConfigNew is empty, so provenance is not captured.
Indexes	Triplicate FULLTEXT indexes on TvSTT and TvVTT are redundant; a single composite index is sufficient.
Data Types	TStamp is stored as text in TvSTT/TvVTT; native temporal types are preferred for correctness and performance.
Constraints	Foreign-keyed columns require indexes; plan for this when introducing ConfigID references.
Flags	Use boolean semantics for channel/format and operational flags.
Normalization	Multi-valued fields (e.g., Categories, keywords) are better modeled as child rows to meet 1NF.

10. Recommendations Summary

- TvSTT:
  - Remove duplicate FULLTEXT indexes; keep only one on the transcript fields (e.g., TEXTS with related text columns).
  - Add a numeric ConfigID foreign key linked to the canonical configuration table to replace text-based provenance.

    - Convert TStamp to a native temporal type (e.g., DATETIME) and use it for time-range filtering.
    - Apply NOT NULL on essential fields such as FName and TStamp to improve data quality.

- TvVTT:
    - Remove duplicate FULLTEXT indexes; keep only one composite index to support caption text search.
    - Add a numeric ConfigID foreign key to link each subtitle record to its configuration.
    - Convert TStamp to a native temporal type and use it for range queries and ordering.
    - Apply NOT NULL on FName and TStamp to enforce completeness.

- TVConfigNew:
    - Adopt a single master configuration table for television capture and make it the authoritative source.
    - Add an is_active (or version) flag to distinguish test versus production configurations.
    - Create a new integer ConfigID (AUTO_INCREMENT PRIMARY KEY) for relational mapping and simpler joins.
    - Normalize repeating station columns (e.g., Station1–Station4) into a child table to satisfy 1NF.

- updated_Email_Alert:
    - Add selective indexes for common filters (recipient email, and date window startdate/enddate) to speed lookups.
    - Store channel/format flags using boolean semantics (MySQL BOOL/BOOLEAN are synonyms for TINYINT(1)).
    - MySQL Developer Zone
    - Split keywords_string into a child table (one keyword per row) to eliminate multi-valued fields (1NF).

- Performance:
    - Drop redundant/unused indexes after normalization and keep one FULLTEXT index per clip table; FULLTEXT applies only to text columns and is used with MATCH … AGAINST.

11. Conclusion

The television tables are organized around capture, content, and alerting, but they currently lack a unifying configuration key and contain redundant text indexes. Consolidating on one FULLTEXT per clip table, adopting native temporal types for timestamps, and introducing a surrogate ConfigID for TvSTT/TvVTT restores referential integrity and reduces write overhead. These changes align with MySQL guidance on FULLTEXT usage, temporal data types, boolean semantics, and foreign-key/index requirements, and they position the subsystem for faster, simpler, and more reliable queries.